
Lecture Notes

on Allometric or Power Laws, Rescaling, and Log Plots

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1 Power Law or Allometry

1.1 Definition:

A **power law function** (also called an **allometric function** in biology) is a function of the form

$$y = ax^k,$$

where

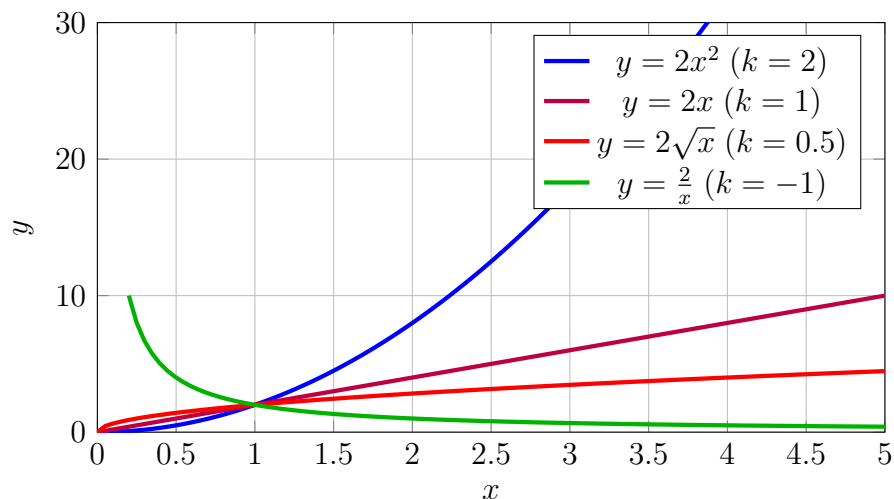
- $a > 0$ is a constant of proportionality, *(dimensionless, no unit)*
- $k \in \mathbb{R}$ is the power (or scaling exponent),
- $x > 0$ is the independent variable.

We will say y is allometric function of x . Sometimes it is also called that x and y are allometrically related.

1.2 Properties:

- If $k > 1$, the function grows faster than linear (*superlinear growth*).
- If $0 < k < 1$, the function grows slower than linear (*sublinear growth*).
- If $k = 1$, the function reduces to a linear function $y = ax$.
- If $k < 0$, the function represents a decreasing relationship, such as inverse proportionality.

Examples of Power Law Functions with $a = 2$



1.3 Real-World Examples of Power Laws (Allometry)

- **Biology (Allometry):** The metabolic rate of animals often scales as body mass to the power of $3/4$, i.e., $B = aM^{3/4}$.
- **Physics:** Gravitational force follows an inverse-square law, $F = \frac{Gm_1m_2}{r^2}$.
- **Economics:** Wealth distributions often follow a Pareto power law.
- **Engineering:** Stress or fracture strength scaling with material size.

Example: Elephant Surface Area (Allometry)

It has been determined that for any elephant, the surface area of the body can be estimated as an allometric function of trunk length. For African elephants, the allometric exponent is 0.74. If a particular elephant has a surface area of 200 ft² and a trunk length of 6 ft, what is the expected surface area of an elephant with a trunk length of 7 ft?

Let S denotes the surface area and L denotes the trunk length.

Let the allometric eqnⁿ (equation) be

$$S = a L^{0.74} \quad \text{--- (*)}$$

*} Given, $S = 200 \text{ ft}^2$ when $L = 6 \text{ ft}$.

$$\therefore (*) \Rightarrow 200 = a (6)^{0.74} \Rightarrow a = \frac{200}{6^{0.74}} = 53.11$$

*} $S(L=7) = ?$

$$\therefore S|_{L=7} = S(L=7) = (53.11) 7^{0.74} = 224.16 \text{ ft}^2.$$

2 Rescaling Data

For x and y as biological variables:

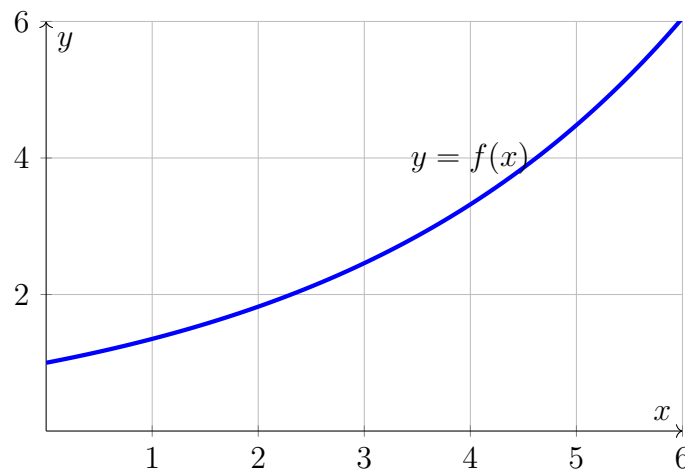
2.1 Definition (Log-Log Graph):

If the horizontal axis is labeled as $\ln(x)$ and the vertical axis is labeled as $\ln(y)$, then the graph is called a **log-log graph**.

2.2 Definition (Semi-Log Graph):

If the horizontal axis is labeled as x and the vertical axis is labeled as $\ln(y)$, then the graph is called a **semi-log graph**.

Example of Semilog Plot



Note: Rescaling data using log or semilog axes is particularly useful for:

- **Exponential functions**, since they appear as **straight lines** on a **semi-log plot**.
- **Power-law (allometric) functions**, since they appear as **straight lines** on a **log-log plot**.
- Consider $f(x) = ab^x$ and $g(x) = cx^k$. Now take the natural logarithm of both sides of each function. What do you obtain?

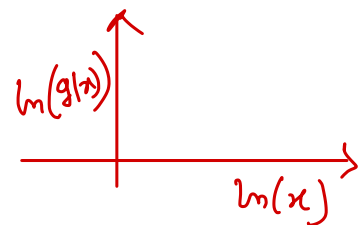
$$\begin{aligned} f(x) &= ab^x \\ \Rightarrow \ln(f(x)) &= \ln(ab^x) \\ \Rightarrow \ln(f(x)) &= \ln(a) + \ln(b^x) \\ \Rightarrow \ln(f(x)) &= \ln(a) + x \ln(b) \end{aligned}$$

$$\text{Form } y = mx + b$$



$$\begin{aligned} g(x) &= cx^k \\ \Rightarrow \ln(g(x)) &= \ln(cx^k) \\ \Rightarrow \ln(g(x)) &= \ln(c) + \ln(x^k) \\ \Rightarrow \ln(g(x)) &= \ln(c) + k \ln(x) \end{aligned}$$

Form $h(x) = a + k \ln(x)$



3 Examples: Rescaling Data

3.1 Exponential Function (Semi-Log Plot)

Consider the exponential function

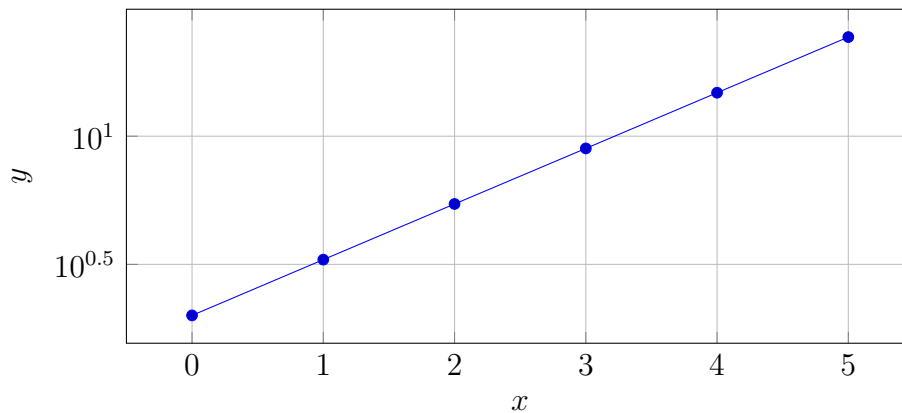
$$y = 2e^{0.5x}, \quad x = 0, 1, 2, 3, 4, 5.$$

The data values are shown below:

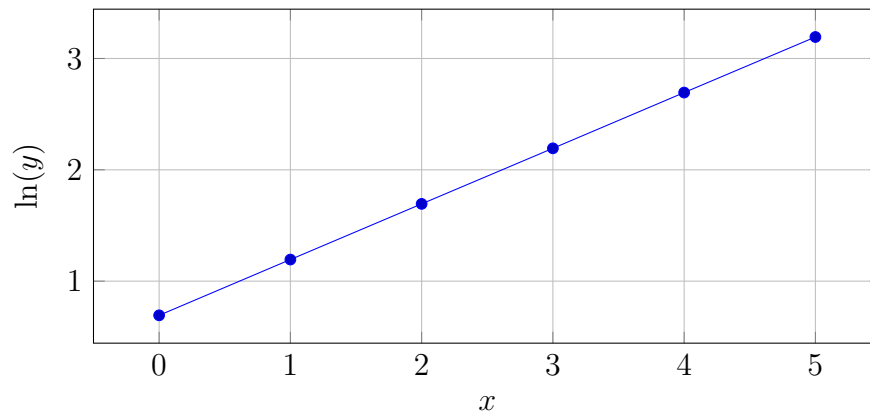
x	$y = 2e^{0.5x}$
0	2.00
1	3.30
2	5.44
3	8.96
4	14.78
5	24.36

Observation: On ordinary axes, the curve rises exponentially. On a **semi-log plot**, the points fall on a straight line.

Exponential Function on Semi-Log Plot



Exponential Function on Semi-Log Plot



3.2 Allometric Function (Log-Log Plot)

Consider the allometric (power-law) function

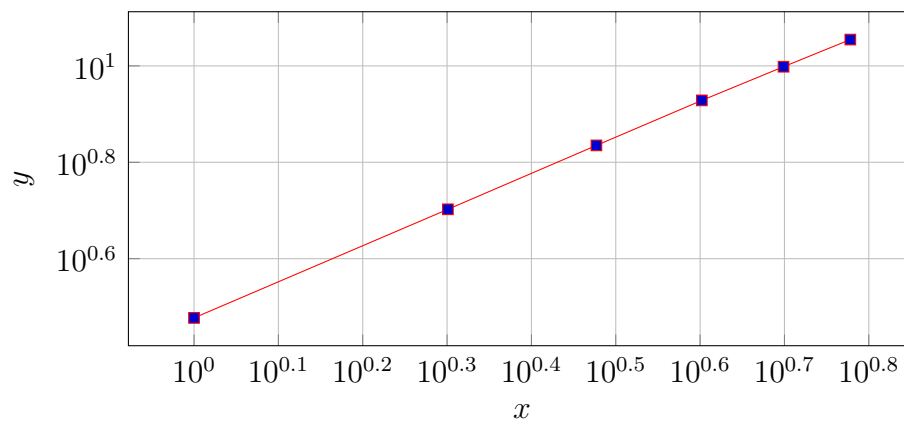
$$y = 3x^{0.75}, \quad x = 1, 2, 3, 4, 5, 6.$$

The data values are shown below:

x	$y = 3x^{0.75}$
1	3.00
2	5.04
3	6.84
4	8.48
5	9.96
6	11.34

Observation: On ordinary axes, the curve increases sublinearly. On a **log-log plot**, the points fall on a straight line with slope 0.75.

Allometric Function on Log-Log Plot



Allometric Function on Log-Log Plot

